

LIDAR Inspection Report of **Box Culvert** ch.932+073

Bridge No: Box Culvert

Bridge Chainage: 932+073(CKRP)

Latitude/Longitude: 25.043812/74.836619



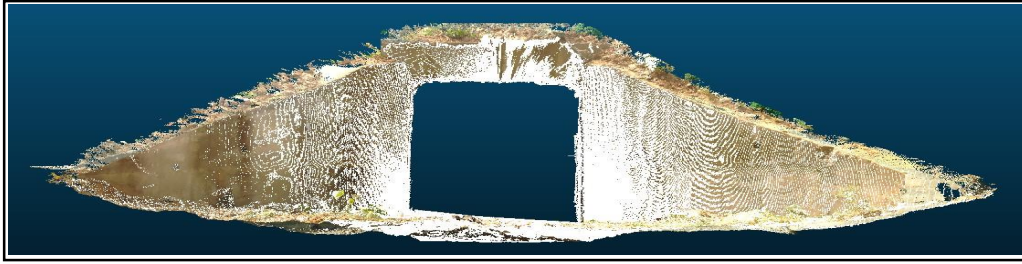
➤ **Bridge Details**

- **Bridge No:** -
- **Chainage:** 932+073
- **Structure type:** Box Culvert

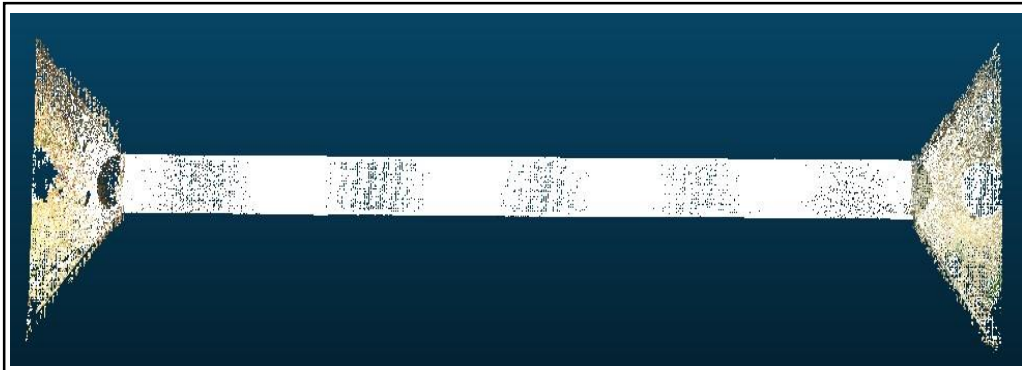
➤ **Project Details**

- **CKRP:** Chittorgarh to kota
- **State:** Rajasthan NH 27
- **Zone:** RJ-06

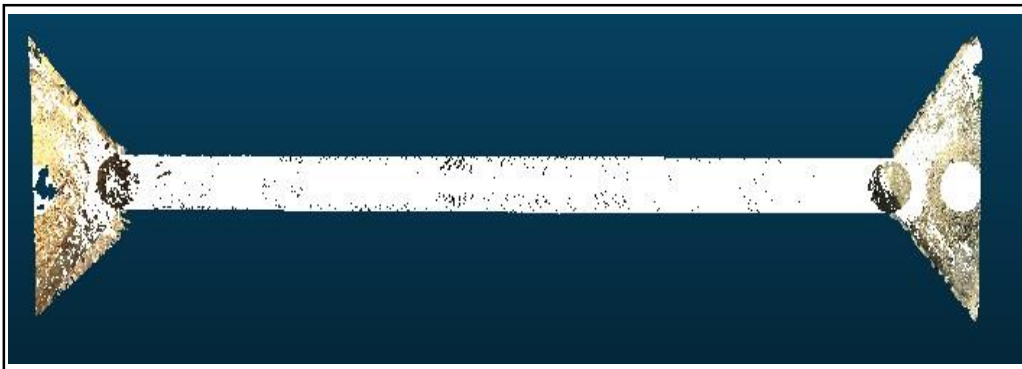
LiDAR View:



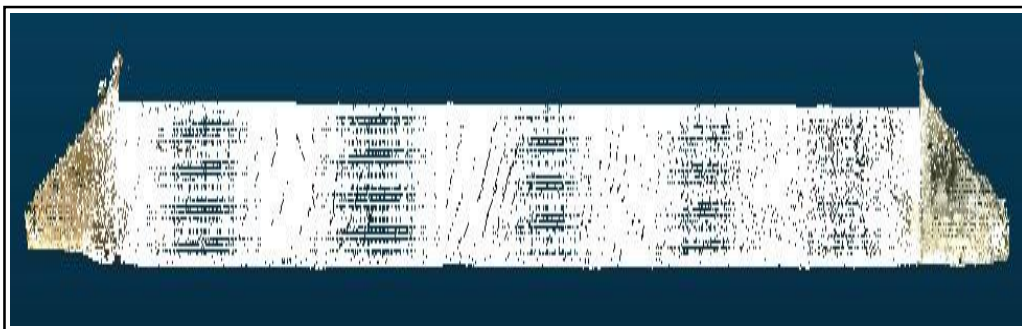
Side View



Top View



Bottom View



Front View

Structural Measurements

1. Height

Measurement	Value
Ground level to slab	2.56 m

2. Length

	Abutment to Abutment
LHS	1.93 m
RHS	1.93 m

Technical Scope of LiDAR Inspection

Category	Details
1. Accuracy Setup and Scan Planning	
Equipment Deployed	Trimble X9 – High-performance terrestrial laser scanner with automatic calibration, IP65-rated rugged build, and in-field registration capability
Point Cloud Density	Configured for high-density capture with sub-centimeter resolution suitable for fine structural detailing
Measurement Accuracy	±1 mm at close range (<60 m); optimal for bridge spans, elevated members, and fascia with scan range up to 150 meters
Coverage Strategy	Comprehensive planning to prioritize critical structural components including abutments, piers, bearings, and deck; ensured overlap to minimize occlusion
2. Field Setup and Scanning Approach	
Site Configuration	Multiple scan positions arranged for 360° coverage including underside views; referenced globally using GNSS/RTK and Ground Control Points (GCPs)
Execution Environment	Scans executed during low-traffic, clear weather conditions to minimize external noise and vibration-related scan distortion
Safety Measures	Operational safety ensured by advance site clearance, barricading, and coordination with local authorities for traffic management
3. Data Resolution and Quality Control	
Resolution Application	Utilized Trimble X9's high-resolution scan settings for capturing large-span structures, vertical components, and minor surface irregularities
Gap Management	Conducted supplementary scans from alternate angles; point cloud refinement and merging handled in Trimble RealWorks and Trimble Perspective
Data Validation	Outlier filtering and noise reduction applied; registration error maintained <5 mm RMSE for consistent dimensional reliability

Equipment Specifications:

Feature	Trimble X9
Scan Speed	1,000,000 pts/sec
Time to Complete a Full Scan	4-6 minutes per setup (depending on resolution)
Point Cloud Accuracy	±3 mm at 20m
Maximum Scanning Range	150m
Field Workflow	Auto-calibration
Imaging	Standard HDR
Durability	IP55

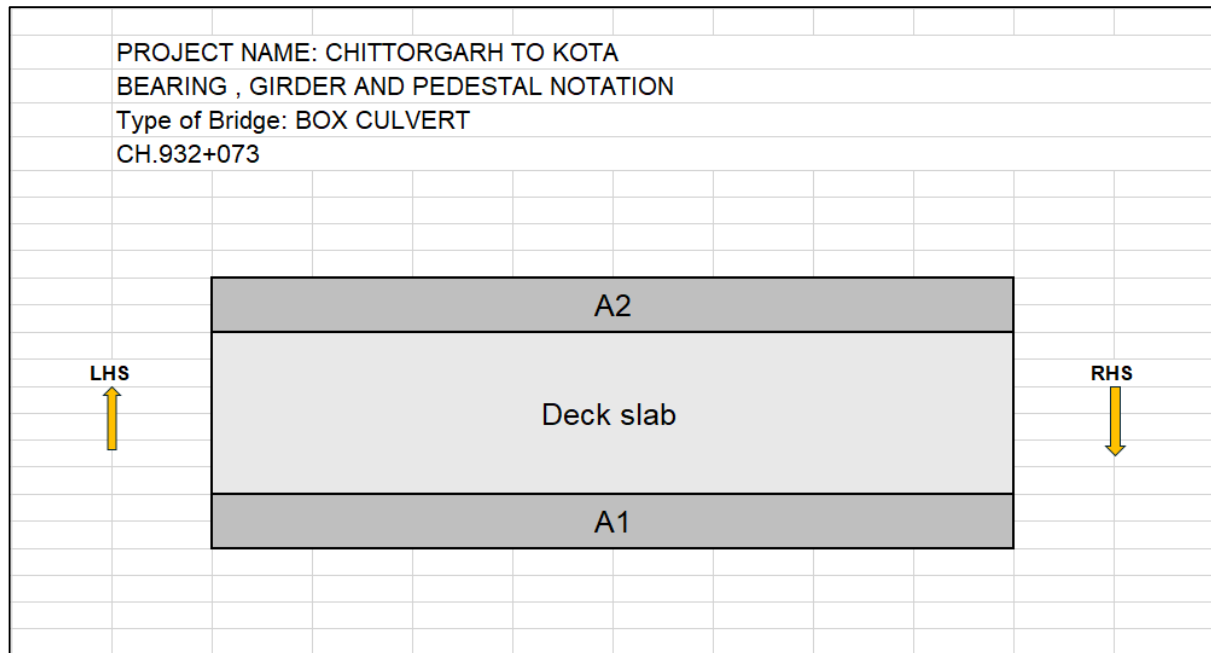
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0. Phase 1 – Scope of Deliverables

Work Stage	Scope / Deliverables	Location in Report
Accuracy Setup & Scan Planning	Specification of accuracy levels, point cloud density, and scan coverage	Technical Scope of LiDAR Inspection
Technical Setup	Preparation for LiDAR equipment deployment	Technical Scope of LiDAR Inspection
Objective Framing	Component prioritization for scanning	Structural Measurements + Observation Sheet
Field Data Collection	LiDAR data acquisition	Structure Layout + Photos
Resolution Analysis	Data density and resolution analysis	Technical Scope of LiDAR Inspection
Gap Filling	Quality enhancement of point cloud	Embedded in Visual/Photo Section
Point Cloud Analysis	Structural details and interpretation	Observation Sheet + Distress Photos

1. Structure Layout



2. Layout Legends

S. No	Description	Notation
1	Abutment	AB
2	Foundation of Abutment	AF
3	Abutment Cap	AC
4	Bearing Pedestal	BP
5	Bearings	BR
6	Foundation of Pier	PF
7	Pier	PR
8	Pier Cap	PC
9	Span	SP
10	Slab span	SL
11	Box Girder span	BG
12	I Girder	IG
13	Rectangular Girder	RG
14	Square Girder	SG
15	T Beam Girders	TB
16	Concrete Deck	CC
17	Bituminous Deck	BD
18	Slab b/n girders	BS
19	Cross Girders	CG
20	Railing LHS	RL
21	Railing RHS	RR
22	Footpath LHS	FL
23	Footpath RHS	RR
24	Expansion Joints	EJ
25	Pourable/Crack Joint Seal	PJ
26	Compression Joint Seal	CJ
27	Assembly Joint Seal	AJ
28	Strip Seal Joint	SJ
29	Drainage Spouts	DS
30	Wing Wall	WW
31	Reinforced earth wall	RE
32	Approach Slab	AS

3. Distress Nomenclature

Legend	Description	Legend	Description
BG	Bulging	SC	Steel Corroded
CRA	Map Crack/ Crazing	SE	Steel Exposed
RC	Crack along main reinforcement	SPE	Seepage Evident
RP	Crack perpendicular to Main Reinforcement	SPM	Seepage Marks
		SL	Scaling
MC	Minor Crack	SG/(G)(S)	Sag in Girder, Slab
WC	Wide Crack	EFF	Efflorescence
SPL	Spalling	STL	Settlement
DL	Delaminated Section	H	Hollow Section
HC	Honey Combing	VEG	Vegetation Growth
TL	Tilt	PD	Pedestal Damage

4. Observations

Element Name	Observation	Location of Distress	Photo No.	Units	Distress Details			Area of Repair (Sq.m/ RMT/ m3)
					L(M)	B(M)	D(M)	
Wing Wall	Damage	-	1	M	0.13	1.26		0.1638
Wing Wall	Damage	-	2	M	0.12	0.62		0.0744
Wing Wall	Damage	-	3	M	0.15	0.76		0.114
Wing Wall	Damage	-	4	M	0.49	0.35	0.04	0.0068
Wing Wall	Damage	-	5	M	0.11	0.69		0.0759

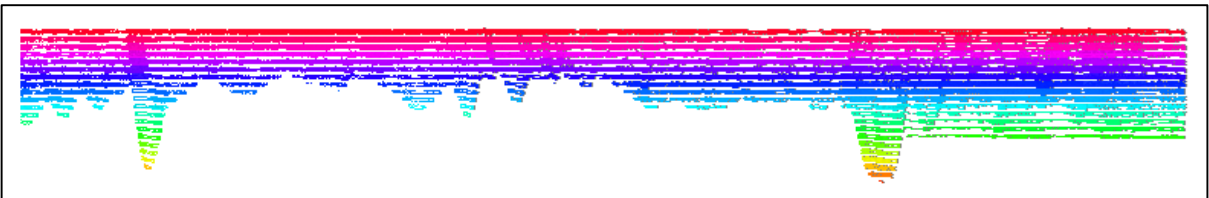
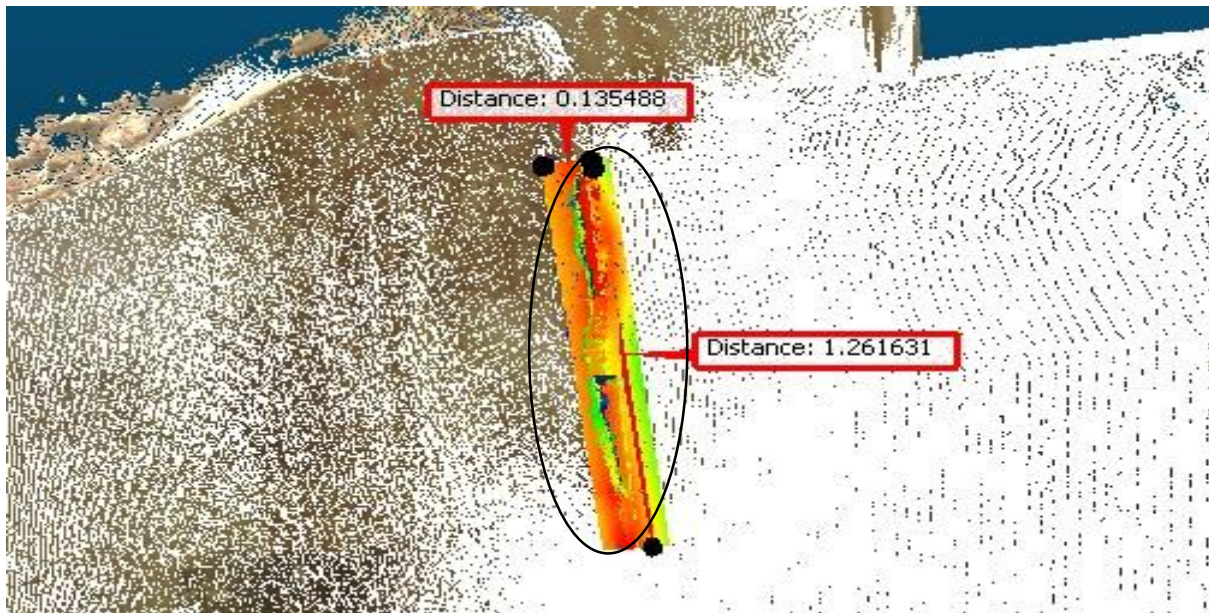
Observation sheet Non- Structural Elements

Sl. No.	Description	Distress Notation	Observation	Location		Distress Details			Area of Repair
				X (m)	Y (m)	L (m)	W (m)	D (m)	Sqm
1	Return wall		OK						
2	Riding return wall		NA						
3	Wing wall		Damage						
4	Butterfly walls at top of median wall		NA						
5	Dirt wall at top of abutment		Ok						
6	Gabion walls		NA						
7	Hazard Marker sign board		OK						
8	Hand railing		NA						
9	Crash barrier		NA						
10	Median wall		NA						
11	Pipe Hand Railing (on crash Barrier)		OK						
12	Cleaning of expansion joints		NA						
13	Wheel guard (Mounted & Unmounted)		NA						
14	Footpath		NA						
15	Drainage Spouts		NA						
16	Drainage Spout Grating		NA						
17	Drainage Spout Extension Pipe		NA						
18	Weep Holes		NA						
19	Fascia Panel		NA						

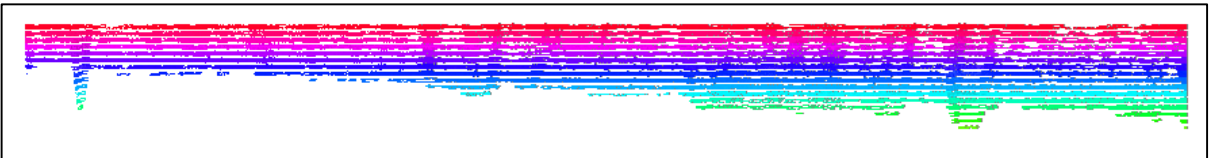
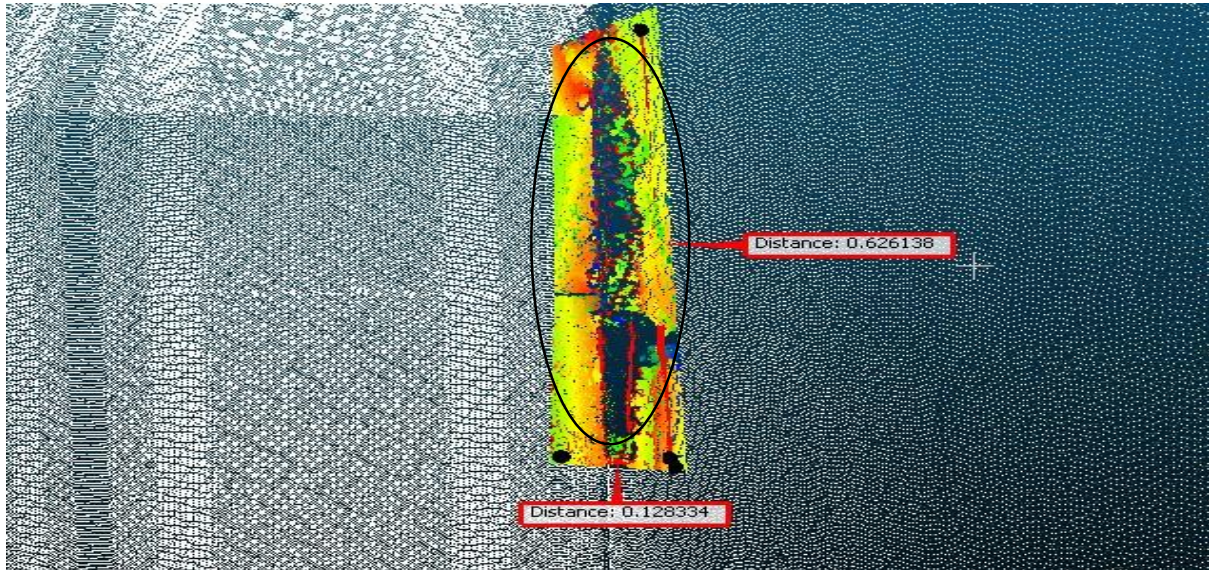
Sl. No.	Description	Distress Notation	Observation	Location		Distress Details			Area of Repair
				X (m)	Y (m)	L (m)	W (m)	D (m)	Sqm
20	Friction Slab		Ok						
21	Approach slab including PCC		OK						
22	Backfilling below Approach slab		OK						
23	Stone Pitching in quarter coning		NA						
24	Filling in quarter coning		NA						
25	Rigid apron		NA						
26	Curtain wall		NA						
27	Flexible apron		NA						
28	Vent clearance		NA						
29	Toe wall of Quarter coning		NA						
30	RCC drains in tunnel		NA						
31	Cover slab of RCC drains		NA						
32	RCC portal of tunnel		NA						
33	Counter fort wall		NA						
34	Booths structure		NA						
35	RCC Booth guard		NA						
36	C.C Panels including PCC		NA						
37	Pipe drain		NA						
38	Pipe drain cover		NA						
39	Drain openings		NA						

CH: 932+073 – Box CULVERT

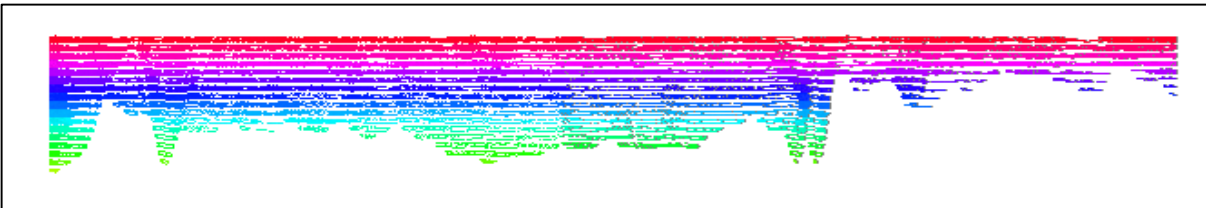
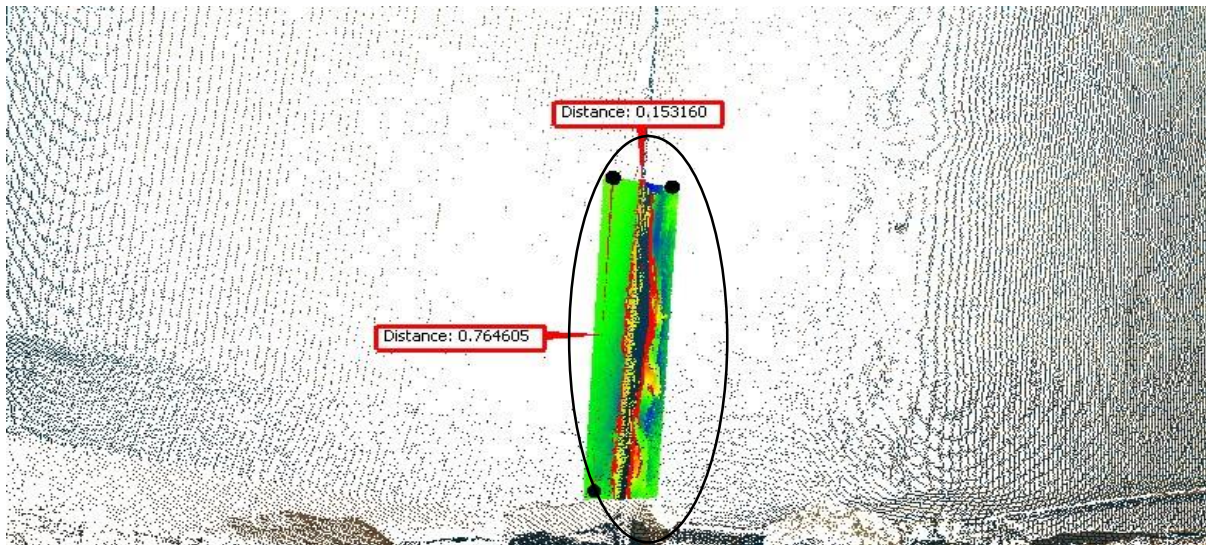
DISTRESS PHOTOS



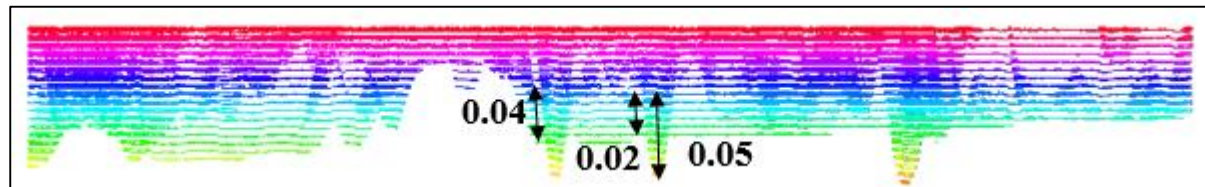
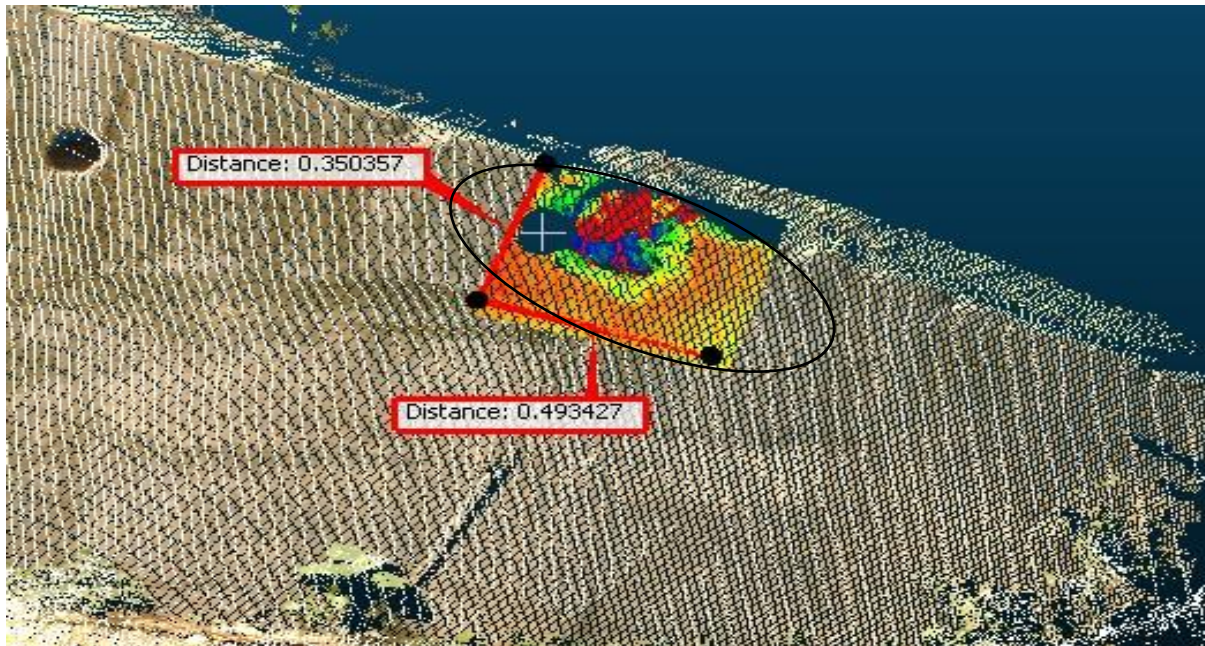
1. "Wing wall damage observed, measuring 0.13 m in length and 1.26 m in width, covering an area of 0.1638 m²."



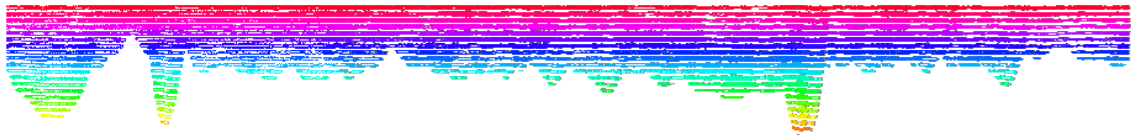
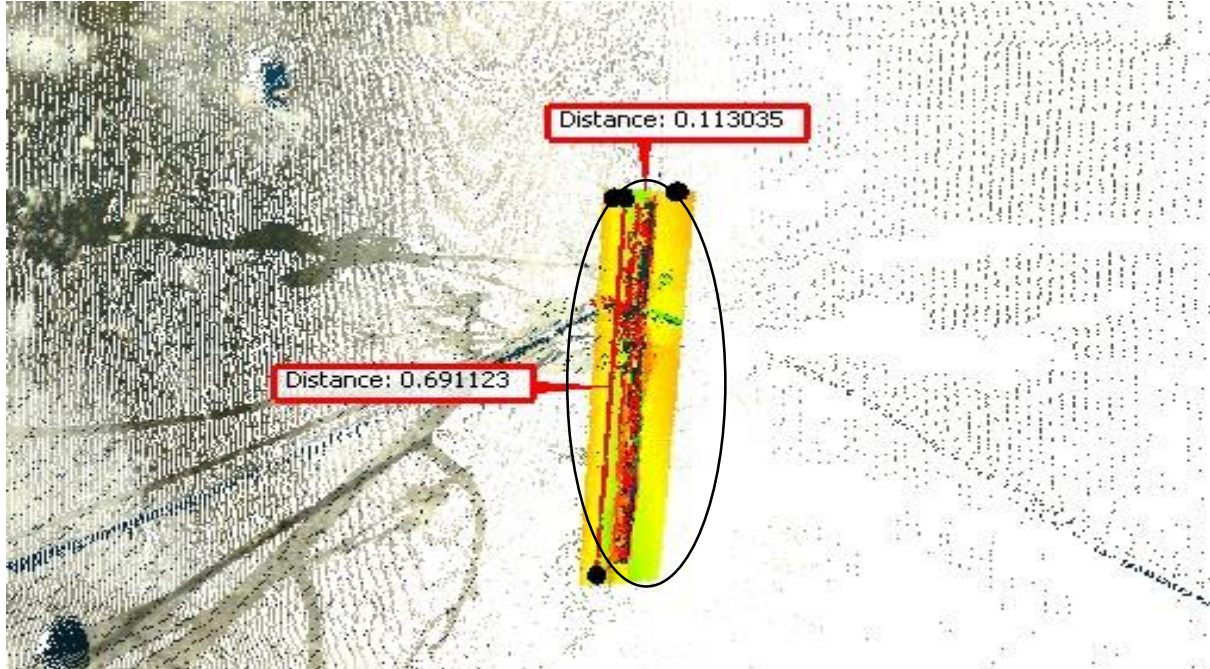
2. "Wing wall damage observed, measuring 0.12 m in length and 0.62 m in width, covering an area of 0.0744 m²."



3. "Wing wall damage observed, measuring 0.15 m in length and 0.76 m in width, covering an area of 0.114 m²."



4. "Wing wall damage observed, measuring 0.49 m in length and 0.35 m in width, Depth ranges from 0.02 m (min) to 0.05 m (max), with an average depth of 0.04 m. The calculated volume of the distress is 0.0068 m³.



5. "Wing wall damage observed, measuring 0.11 m in length and 0.69 m in width, covering an area of 0.0759 m²."